

Do Histology–Transcriptomics Survival Models Use Both Inputs? A Missing-Modality Audit and a One-Checkpoint Training Recipe

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Abstract

Survival models that fuse whole-slide images (WSIs) with transcriptomics are evaluated on patients with both inputs, leaving two questions open: does the fused model use both, and what happens when one is missing? We audit this on five TCGA cohorts under one equal-budget, three-seed protocol by re-evaluating checkpoints with an input removed or permuted across patients. The official SurvPath implementation ranks patients the same way when its RNA input is permuted (Δ C-index +0.001) but loses 0.068 when the slide is permuted and falls to 0.47–0.52 without it. A query-bottleneck fusion model trained with null codes and modality dropout is equally blind to RNA removal but loses only 0.030 without the slide; auxiliary unimodal heads make the fused prediction respond to RNA (−0.023 under permutation), and one checkpoint keeps a C-index of 0.54–0.62 with either input removed and moves at most 0.025 when half the patients lack a modality. With both inputs it does not detectably beat an RNA-only MLP or a late-fusion ensemble, and its +0.036 margin over the untuned SurvPath is significant under the Wilcoxon test but not the t -test. Replayed on stored histories, validation-loss early stopping would have cost it up to 0.09 C-index and SurvPath nothing.

1 Introduction

Prognostic models in oncology draw on two kinds of evidence: histology records the phenotype of a tumour, and transcriptomics records the molecular state behind it. Pathology foundation models (Chen et al., 2024; Xu et al., 2024; Vorontsov et al., 2024) turn a gigapixel whole-slide image (WSI) into a few thousand frozen patch embeddings, and co-attention between patch tokens and gene or pathway tokens (Chen et al., 2021; Jaume et al., 2024) relates the two sets, so models that fuse WSIs with bulk gene expression report higher concordance than either modality alone on most cohorts of The Cancer Genome Atlas (TCGA) (Chen et al., 2022b; Jaume et al., 2024; Song et al., 2024).¹

Such models are evaluated on patients who have both inputs, which leaves two questions open. Does the fused model use both inputs? A concordance index (C-index) measures how well predicted risks rank patients; it cannot show which input produced the ranking. And what happens when one input is missing, the normal state of a hospital cohort, where sequencing is not ordered for every patient and not every slide is digitised? We answer both with an input-use audit: every trained checkpoint is re-evaluated with one input removed, with one input permuted across patients, and with a random fraction of patients lacking a modality, and its fused risk scores are correlated with the risks of its own single-input evaluations. Applied under one protocol to the official SurvPath implementation (Jaume et al., 2024), the audit finds that SurvPath’s ranking does not respond to its RNA input (Δ C-index +0.001 under permutation across patients) and collapses when the slide is permuted (Δ C-index −0.068) or removed (C-index 0.47–0.52).

¹A large language model assisted with drafting and editing the text and L^AT_EX source and with writing analysis and plotting code, under the authors’ direction. The authors designed the study, ran the experiments, checked every reported number against the archived results, and are responsible for all content.

The second part of the paper asks which training choices change this. We train one query-bottleneck fusion model, PathQ-Former, in two variants that differ only in training. With learned null codes and modality dropout alone, the model is as blind to RNA as SurvPath (Δ C-index 0.000 with RNA removed) but no longer collapses without the slide (-0.030 instead of SurvPath’s -0.073). Adding auxiliary unimodal survival heads makes the fused prediction respond to RNA (-0.023 under permutation); the resulting checkpoint keeps a C-index of 0.54–0.62 with either input removed and changes by at most 0.025 when half of the patients lack a modality. With both inputs present, neither variant is detectably better than an RNA-only multilayer perceptron (MLP) or a late-fusion ensemble of two unimodal models, so the recipe buys coverage of incomplete patients, not accuracy. Our contributions are:

- An input-use audit (removal, permutation across patients, risk-score correlation; 3 seeds $\times$ 5 folds $\times$ 5 cohorts) applied to a published co-attention model under an equal-budget protocol (Section 4.1). SurvPath’s ranking does not respond to its RNA input; a high fused C-index does not show that both modalities are used.
- Evidence on where missing-modality robustness comes from, from matched variants of one fusion model (Sections 4.3–4.4): null codes and modality dropout remove the collapse without the slide (-0.030 with the null code against -0.069 with mean imputation on the same checkpoints, one seed, and -0.073 for SurvPath); auxiliary unimodal heads are what makes the fused output respond to RNA; the single checkpoint matches an RNA-only MLP and a late-fusion ensemble with both inputs and works with either input absent.
- Protocol evidence that this literature rarely reports (Sections 4.2 and 4.5): equal-budget accuracy with seed-aware statistics (seed-averaged 25-fold paired tests with bootstrap confidence intervals and Holm correction), and a post-hoc replay of checkpoint selection on event-poor folds from the stored training histories.

Scope of claims SurvPath was run with its published hyper-parameters and was not re-tuned under our protocol. PathQ-Former’s hyper-parameters were developed on the bladder cohort (BLCA) under a 10-epoch budget, and the auxiliary-head variant and its loss weight were adopted after the 20-epoch results on all five cohorts had been seen, so both variants are reported as co-primary and no accuracy superiority over SurvPath is claimed. The training recipe has not been applied to SurvPath. All results are on TCGA with frozen UNI2-h features, simulated missingness is random and therefore uninformative, and no clinical covariates are modelled.

2 Related Work

Fusion of WSIs and genomics WSIs are processed as bags of patches under multiple-instance learning: attention-based MIL (ABMIL) (Ilse et al., 2018), clustering-constrained attention MIL (CLAM) (Lu et al., 2021) and TransMIL (Shao et al., 2021) pool patch features from self-supervised (Wang et al., 2022) or foundation-model encoders trained on 10^8 patches or more (Chen et al., 2024; Xu et al., 2024; Vorontsov et al., 2024). Early fusion work concatenated convolutional neural network (CNN) and genomic features (Mobadersany et al., 2018) or combined unimodal representations with Kronecker products and gating (Chen et al., 2022a); a pan-cancer study found that fusion improved concordance in 12 of 14 cancer types (Chen et al., 2022b). MCAT (Chen et al., 2021) introduced co-attention from functional gene groups to patches; SurvPath (Jaume et al., 2024) replaced the groups with Reactome and Hallmark pathway tokens and modelled dense pathway–patch interactions in one Transformer; MOTCat (Xu & Chen, 2023), CMTA (Zhou & Chen, 2023), PIBD (Zhang et al., 2024) and MMP (Song et al., 2024), which reports gains over SurvPath, vary the interaction, and continually evolved multimodal foundation models extend the setting (Peng et al., 2025). Two-bank learned-query fusion over patches and pathways already exists in APL (Liu et al., 2025b); the query bottleneck used here is not claimed as new.

Missing modalities in multimodal survival Modality dropout was introduced for gesture recognition (Neverova et al., 2016) and applied to pan-cancer prognosis by Cheerla & Gevaert (2019) and MultiSurv (Vale-Silva & Rohr, 2021). Ma et al. (2022) showed that multimodal Transformers are not robust

Table 1: Cohorts as realised by the DSS splits (every method and seed uses the same five folds). Patients and events per validation fold are the minimum–maximum over the five folds. [†]Development cohort for PathQ-Former’s hyper-parameters.

Cohort	Site	Patients	DSS events	Patients per fold	Events per fold
BLCA [†]	bladder	359	113	70–75	16–25
BRCA	breast	868	60	166–182	7–20
COADREAD	colon and rectum	296	37	45–71	5–13
HNSC	head and neck	392	117	56–126	13–45
STAD	stomach	318	84	42–101	9–33

to missing inputs by default; SMIL (Ma et al., 2021), ShaSpec (Wang et al., 2023) and missing-aware prompts (Lee et al., 2023) address it with meta-learning, shared-specific features or prompt tokens. For WSIs and genomics, G-HANet (Wang et al., 2025) distils genomic knowledge into a histology-only model, and recent work binds incomplete modalities (Qu et al., 2024), distils prompts (Xu et al., 2025), learns conditional latent variables (Zhou et al., 2025), uses prototype-guided cross-modal knowledge (Liu et al., 2025a), memory-augmented hypergraphs (Qu et al., 2025) or prototype-driven pretraining (Yu et al., 2026). None of these audits whether a published co-attention model uses its genomic input, nor tests under equal budgets whether unimodal auxiliary supervision changes that; this is the gap we address.

Learned queries A set of learned vectors that cross-attend to a large input and return a fixed-size summary appears as inducing points (Lee et al., 2019), object queries (Carion et al., 2020), the Perceiver latent array (Jaegle et al., 2021), the Perceiver Resampler (Alayrac et al., 2022), the Q-Former (Li et al., 2023) and the prototype banks of APL (Liu et al., 2025b). The property used here is that the output has a fixed shape, so an absent input can be replaced by a learned tensor of the same shape.

Modality imbalance Gradient Blending (Wang et al., 2020) and OGM-GE (Peng et al., 2022) showed that joint training lets one modality dominate and counteract it with per-modality supervision or gradient modulation. The auxiliary unimodal heads used here are unimodal supervision in that tradition, not a new idea; our contribution is to measure what they change in a survival-fusion model under an audit.

3 Experimental Setup

3.1 Data and cohorts

Five TCGA cohorts are used (Table 1): bladder urothelial carcinoma (BLCA), breast invasive carcinoma (BRCA), colorectal adenocarcinoma (COADREAD), head and neck squamous cell carcinoma (HNSC) and stomach adenocarcinoma (STAD). Patch embeddings are the released 1,536-d UNI2-h features (Chen et al., 2024; Mahmood Lab, 2025a;b), used as released. The RNA-seq matrices (4,999 genes), clinical metadata and five-fold patient-level splits are those released with SurvPath (Jaume et al., 2024). The primary endpoint is disease-specific survival (DSS) from the TCGA Clinical Data Resource (Liu et al., 2018); overall survival (OS) is secondary. The split is fixed per cohort, so fold k holds the same validation patients for every method and seed, and the number of events in a validation fold ranges from 5 to 45 (median 16).

3.2 Protocol

Table 2 states the protocol, which is applied without change to every method. Every method trains for a fixed budget of 20 epochs and reports the final checkpoint, the convention of MCAT and SurvPath; no checkpoint is selected on the validation fold, which is also the reported fold. The budget was set after inspecting pooled validation curves under a 10-epoch budget (Appendix A), so 10-epoch results are reported as well. Appendix F lists evaluation pitfalls of survival MIL pipelines that the protocol avoids.

Table 2: Evaluation protocol, identical for all methods.

Aspect	Choice
Unit of analysis	patient; all slides of a patient concatenated
Splits	SurvPath’s five-fold patient-level cross-validation splits; the validation fold is the reported fold
Endpoint	disease-specific survival; overall survival as a secondary endpoint
Time bins	quartiles of uncensored training patients, outer edges $\pm\infty$, reused for validation
Primary metric	Harrell’s C-index on continuous time (Harrell et al., 1982; Pölsterl, 2020)
Secondary metrics	Uno’s inverse-probability-of-censoring-weighted (IPCW) C-index (Uno et al., 2011) truncated at the 75th percentile of training event times; integrated Brier score (Graf et al., 1999) on the interior quartile grid; log-rank test at the median risk split (Appendices B and E)
Training budget	20 epochs, final checkpoint, all methods
Seeds	three (0, 1, 2); the fold index is added to the seed
Statistics	one value per (cohort, fold) = mean over seeds; paired t and Wilcoxon signed-rank tests over the 25 pooled folds with a 95% percentile-bootstrap CI of the mean difference (10,000 resamples of folds) as the primary test; per-cohort tests ($n = 5$) are Holm-adjusted over the five cohorts and exploratory
Audit	the same checkpoint evaluated with each input removed or permuted across patients, with 10–50% of patients missing one modality at random (five draws), and Spearman correlation between its fused and single-input risks

Statistics The three seeds of a fold share its validation patients, so (fold, seed) pairs are not independent replicates (Nadeau & Bengio, 2003). The unit of analysis is therefore the (cohort, fold): the C-index of each method is averaged over the seeds available for both methods, the paired difference is formed on each of the 25 folds, and a paired t -test, a two-sided exact Wilcoxon signed-rank test and a 95% percentile-bootstrap confidence interval (CI) of the mean difference are computed over the folds. Per-cohort tests have five folds each; their p -values are Holm-adjusted over the five cohorts and are exploratory. “Mean $\pm$ sd” in a table is the mean and standard deviation over seeds of the five-fold mean unless the caption says otherwise.

3.3 Methods

Table 3 lists the methods and the seeds run for each. SurvPath, ABMIL, SNN and the RNA MLP use the hyper-parameters of the SurvPath repository; none was tuned under this protocol.

PathQ-Former Figure 1 shows the model; equations and hyper-parameters are in Appendix A, compute cost in Appendix C and the attention over pathway tokens in Appendix D. Patient i has patch embeddings $X_i^h \in \mathbb{R}^{N_i \times 1536}$ from all of the patient’s slides and a gene vector $x_i^g \in \mathbb{R}^{4999}$, min-max scaled to $[-1, 1]$ on the training split. Genes are grouped into $P = 275$ pathway tokens by per-pathway two-layer MLPs, using the Reactome (Gillespie et al., 2022) and MSigDB Hallmark (Liberzon et al., 2015) definitions released with SurvPath (331 pathways) with our filter of 3–300 member genes. Each modality $m \in \{h, g\}$ has $K = 32$ learned queries and two pre-norm blocks of cross-attention to its input tokens, self-attention among the queries and a feed-forward layer, which return a code $Z^m \in \mathbb{R}^{32 \times 256}$. The two codes are concatenated and passed through two Transformer encoder layers (Vaswani et al., 2017); attention pooling and a sigmoid layer give $T = 4$ discrete-time hazards h_t with bins at the quartiles of the uncensored training times, survival $S(t) = \prod_{k \leq t} (1 - h_k)$ and risk $r = -\sum_t S(t)$, as in MCAT and SurvPath (Zadeh & Schmid, 2021). For a patient with bin y and censoring indicator c the loss is the negative log-likelihood

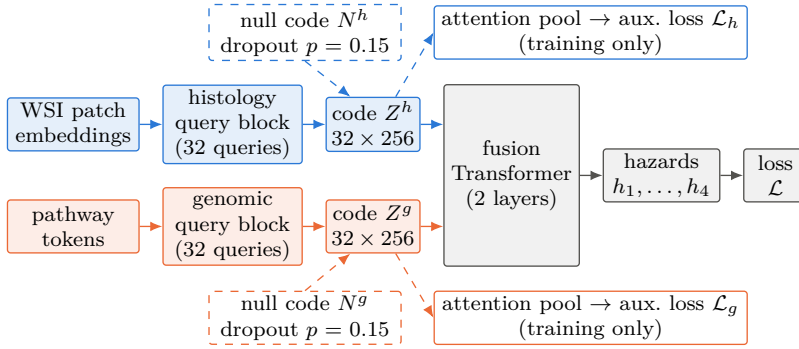
$$\mathcal{L}(y, c) = -(1 - c)[\log S(y - 1) + \log h_y] - c \log S(y), \quad (1)$$

with the censored term unweighted for PathQ-Former and weighted by 0.5 for SurvPath, its published setting. *Training for missingness.* A learned null code $N^m \in \mathbb{R}^{32 \times 256}$ replaces Z^m when modality m is absent. During training each present modality is dropped with probability 0.15 per sample, independently, with at least one modality kept, and the dropped code is replaced by its null code, so the fusion Transformer

Table 3: Methods and runs. “Seeds” is the number of seeds under the 20-epoch DSS protocol; one BRCA run of each single-modality PathQ-Former is missing at seed 2. Parameter counts include the pathway MLPs.

Method	Inputs	Training	Seeds	Also run under
SurvPath (official) (Jaume et al., 2024)	WSI+RNA	published hyper-parameters; 21.2 M params	3	OS (1); 10 epochs (3)
ABMIL (Ilse et al., 2018)	WSI	repository defaults	3	–
SNN (self-normalizing network) (Klambauer et al., 2017)	RNA	repository defaults	3	–
RNA MLP	RNA	two layers, repository defaults	3	–
Slide-only PathQ-Former	WSI	trained with RNA absent	3 (BRCA 2)	–
RNA-only PathQ-Former	RNA	trained with slide absent	3 (BRCA 2)	–
Late fusion	WSI+RNA	mean of the z -scored risks of the slide-only and RNA-only PathQ-Formers (same seed)	2	–
PathQ-Former w/o aux heads	WSI+RNA	null codes, modality dropout; 16.9 M params	3	10 epochs (3)
PathQ-Former	WSI+RNA	as above plus auxiliary heads, $\lambda = 0.5$	3	OS (1)

(a) Training



(b) Inference

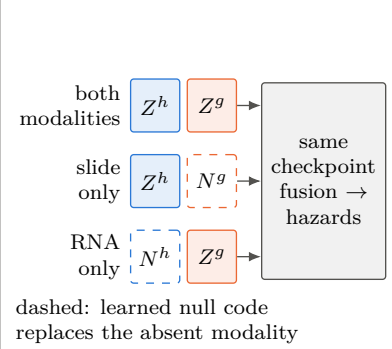


Figure 1: PathQ-Former is trained for missingness and served from one checkpoint. (a) Training. The WSI patch embeddings and the pathway tokens are each read by a block of 32 learned queries, giving codes $Z^h, Z^g \in \mathbb{R}^{32 \times 256}$ that a two-layer fusion Transformer maps to four discrete-time hazards and the survival loss $\mathcal{L}$. With probability $p = 0.15$ per sample, modality dropout replaces a present code by its learned null code N^h or N^g (dashed), and auxiliary survival losses $\mathcal{L}_h$ and $\mathcal{L}_g$ on the attention-pooled codes make each branch prognostic on its own (training only). (b) Inference. Patients with both modalities, a slide only or RNA only enter the same checkpoint; the code of the absent modality is replaced by its null code and nothing else changes.

sees null codes in training in the form it sees them at test time. *Auxiliary heads.* In the full model, attention pooling of Z^h and of Z^g each feeds a linear hazard head trained with Eq. (1) on the samples in which that modality is present; the objective is $\mathcal{L}_{\text{fused}} + \lambda \sum_m \mathbf{1}[m \text{ present}] \mathcal{L}_m$ with $\lambda = 0.5$, and the heads are discarded at test time. “PathQ-Former w/o aux heads” is the same model trained with $\lambda = 0$.

3.4 The audit

Each trained checkpoint is evaluated on its validation fold under five conditions: *Complete*; *RNA removed*; *RNA permuted*, where the gene vectors are shuffled across the validation patients so that every patient receives a real but wrong profile; *Slide removed*; and *Slide permuted*. For PathQ-Former a removed modality is replaced by its null code. SurvPath cannot run without both inputs, so it receives the training-mean gene

Table 4: Input-use audit. C-index of the same 20-epoch checkpoint under five test-time conditions: mean $\pm$ sample standard deviation over three seeds of the five-fold mean (as in Table 5). A removed input is the training mean for SurvPath and the learned null code for PathQ-Former; a permuted input is shuffled across the validation patients. Δ : pooled difference from Complete over the 25 seed-averaged folds, with paired t -test and Wilcoxon p -values. The variant without auxiliary heads has no permuted evaluation (n/a). [†]Development cohort.

Condition	BLCA [†]	BRCA	COADREAD	HNSC	STAD	Δ	t p	W p
<i>SurvPath (official), mean imputation</i>								
Complete	0.594 $\pm$ 0.012	0.536 $\pm$ 0.026	0.570 $\pm$ 0.038	0.552 $\pm$ 0.019	0.587 $\pm$ 0.008			
RNA removed	0.595 $\pm$ 0.013	0.536 $\pm$ 0.027	0.572 $\pm$ 0.037	0.553 $\pm$ 0.018	0.588 $\pm$ 0.007	+0.001	0.12	0.085
RNA permuted	0.594 $\pm$ 0.013	0.537 $\pm$ 0.028	0.572 $\pm$ 0.038	0.552 $\pm$ 0.011	0.587 $\pm$ 0.006	+0.001	0.36	0.13
Slide removed	0.486 $\pm$ 0.022	0.520 $\pm$ 0.025	0.473 $\pm$ 0.012	0.504 $\pm$ 0.043	0.493 $\pm$ 0.082	−0.073	0.003	0.002
Slide permuted	0.511 $\pm$ 0.018	0.532 $\pm$ 0.052	0.510 $\pm$ 0.024	0.489 $\pm$ 0.033	0.459 $\pm$ 0.016	−0.068	0.016	0.007
<i>PathQ-Former w/o aux heads, null codes</i>								
Complete	0.609 $\pm$ 0.016	0.611 $\pm$ 0.039	0.638 $\pm$ 0.031	0.575 $\pm$ 0.012	0.593 $\pm$ 0.015			
RNA removed	0.612 $\pm$ 0.020	0.600 $\pm$ 0.046	0.633 $\pm$ 0.027	0.588 $\pm$ 0.008	0.595 $\pm$ 0.021	+0.000	0.96	0.51
RNA permuted	n/a	n/a	n/a	n/a	n/a			
Slide removed	0.590 $\pm$ 0.050	0.572 $\pm$ 0.033	0.601 $\pm$ 0.056	0.533 $\pm$ 0.017	0.578 $\pm$ 0.035	−0.030	0.080	0.16
Slide permuted	n/a	n/a	n/a	n/a	n/a			
<i>PathQ-Former, null codes</i>								
Complete	0.630 $\pm$ 0.010	0.622 $\pm$ 0.033	0.613 $\pm$ 0.023	0.582 $\pm$ 0.025	0.572 $\pm$ 0.022			
RNA removed	0.600 $\pm$ 0.002	0.592 $\pm$ 0.047	0.576 $\pm$ 0.025	0.596 $\pm$ 0.016	0.589 $\pm$ 0.010	−0.013	0.27	0.43
RNA permuted	0.597 $\pm$ 0.010	0.572 $\pm$ 0.046	0.618 $\pm$ 0.028	0.566 $\pm$ 0.019	0.555 $\pm$ 0.007	−0.023	0.075	0.063
Slide removed	0.613 $\pm$ 0.022	0.617 $\pm$ 0.052	0.583 $\pm$ 0.022	0.545 $\pm$ 0.026	0.538 $\pm$ 0.004	−0.025	0.19	0.11
Slide permuted	0.570 $\pm$ 0.035	0.539 $\pm$ 0.027	0.556 $\pm$ 0.033	0.539 $\pm$ 0.058	0.484 $\pm$ 0.035	−0.067	0.0002	0.0004

vector or the training-mean patch embedding in place of a removed one; PathQ-Former is also evaluated under the same mean imputation, to separate the effect of the null code from that of the architecture. Permutation is the stronger test: a mean or null input is neutral, whereas a permuted input is plausible and wrong, so a model that reads it must lose concordance. In the *random-missingness sweep*, 10, 20, 30 or 50% of validation patients lack one modality at random (five draws per fold, mean reported). Finally, the *risk-score correlation* is the Spearman ρ between a checkpoint’s fused risks and the risks of its own RNA-removed and slide-removed evaluations on the same fold, and between its fused risks and those of the other methods on the same fold and seed. The variant without auxiliary heads has no permuted evaluation.

4 Results

4.1 Do the models use both inputs?

Table 4 is the audit and Figure 2 plots its pooled differences. SurvPath’s C-index is unchanged when its RNA input is removed (+0.001; $p = 0.12$ and 0.085) and when it is permuted across patients (+0.001; $p = 0.36$ and 0.13): the model ranks patients the same way with their own profile, the training mean or another patient’s profile. Permuting the slide costs it 0.068 ($p = 0.016$ and 0.007) and removing the slide 0.073 ($p = 0.003$ and 0.002), leaving 0.47–0.52 (removed) and 0.46–0.53 (permuted) on every cohort; the smallest change is on BRCA (−0.016), where its complete-input score (0.536) is already close to chance. PathQ-Former without auxiliary heads behaves the same way with respect to RNA (RNA removed +0.000; $p = 0.96$ and 0.51) but loses only 0.030 without the slide ($p = 0.080$ and 0.16). With auxiliary heads the fused prediction responds to RNA: −0.013 when it is removed ($p = 0.27$ and 0.43) and −0.023 when it is permuted ($p = 0.075$ and 0.063, short of 0.05 under the seed-averaged test; the attribution rests on this direction and on the contrast with the +0.000 and +0.001 of the other two models), the ordering expected of an input that is read (a wrong profile costs more than a null one); removing the slide costs 0.025 ($p = 0.19$ and 0.11) and permuting it 0.067 ($p = 0.0002$ and 0.0004), and the C-index stays at 0.54–0.62 with either input removed.

The risk-score correlations say how the fused prediction is composed. For PathQ-Former, the Spearman ρ between the fused risk and the same checkpoint’s RNA-removed risk is 0.83 pooled over folds and seeds,

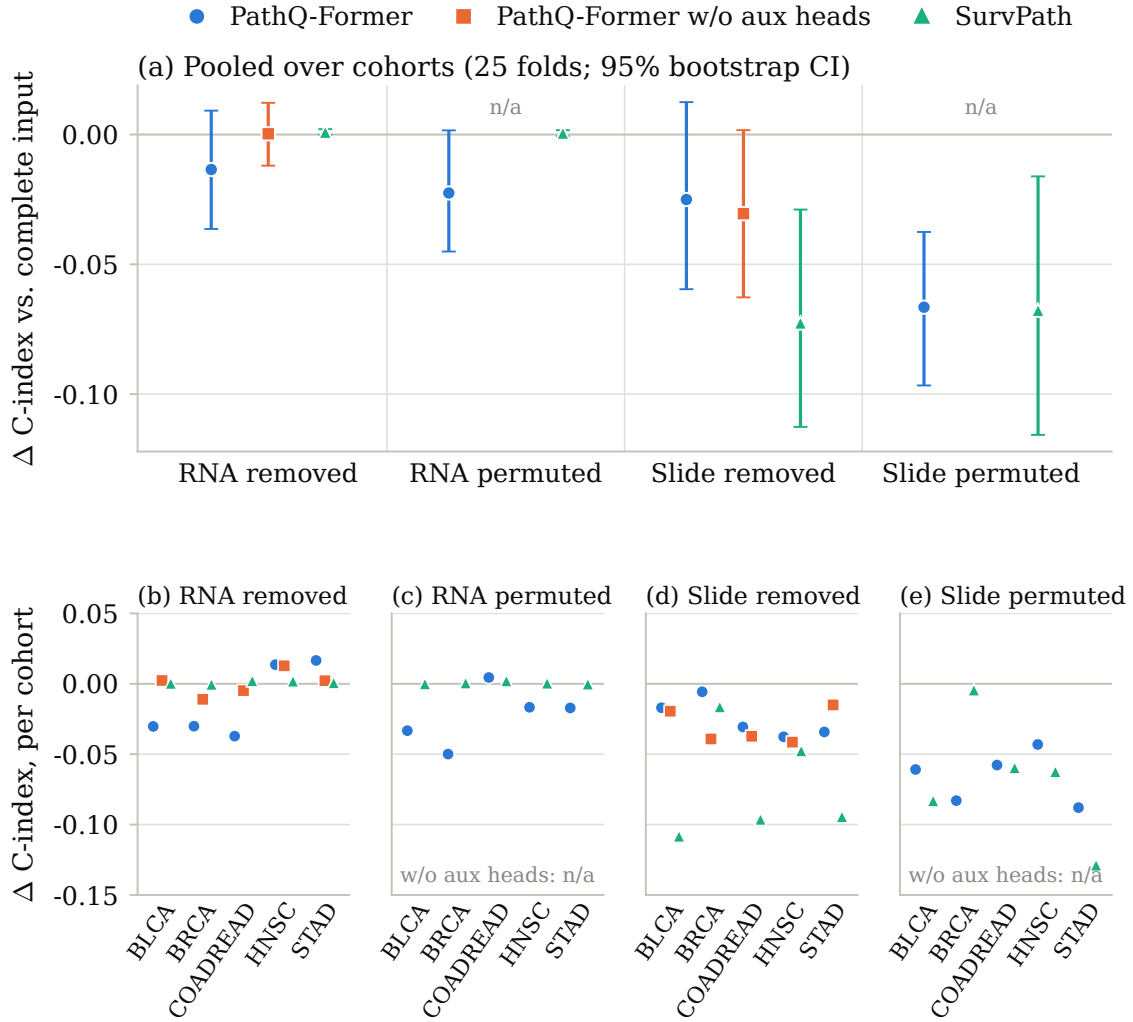


Figure 2: Change in C-index of the same 20-epoch checkpoint when one test-time input is removed or permuted across validation patients, relative to its complete-input evaluation. SurvPath receives the training-mean gene vector or patch embedding for a removed input; PathQ-Former uses its learned null codes. (a) Pooled over the five cohorts: per (cohort, fold) the change is averaged over three seeds, then the mean and 95% bootstrap confidence interval over the 25 folds are shown (10,000 resamples of folds). (b)–(e) The same quantity per cohort (mean over five folds). The variant without auxiliary heads was not evaluated under permutation (n/a).

against 0.45 with its slide-removed risk, and the two single-input risks correlate at 0.15 (Table 13). The fused prediction therefore remains histology-led: the auxiliary heads make it respond to RNA, they do not make the two inputs symmetric. Its correlation with the other models’ risks on the same fold and seed is 0.69 with the variant without auxiliary heads, 0.46 with SurvPath, 0.46 with ABMIL and 0.35 with the RNA MLP.

4.2 Accuracy at equal budget

Table 5 gives the C-index of every method and Table 6 the seed-averaged paired tests. Pooled over 25 folds, PathQ-Former exceeds SurvPath by 0.036 (95% CI $[-0.004, +0.071]$; t -test $p = 0.074$, Wilcoxon $p = 0.030$; 18 of 25 folds) and the variant without auxiliary heads by 0.037 ($[+0.009, +0.065]$; $p = 0.020$ and 0.020 ; 17 of 25). Per cohort, BRCA is the only cohort with a raw $p < 0.05$ ($+0.086$, t -test $p = 0.017$), and it does not survive Holm correction over the five cohorts ($p = 0.084$); no DSS cohort survives correction (Table 9).

Table 5: Disease-specific survival C-index under the 20-epoch protocol: mean $\pm$ sample standard deviation over seeds of the five-fold mean (three seeds; late fusion two seeds; the BRCA cells of the single-modality PathQ-Formers two seeds). Pooled is the mean of the five cohort means. [†]Development cohort.

Method	Inputs	BLCA [†]	BRCA	COADREAD	HNSC	STAD	Pooled
PathQ-Former	WSI+RNA	0.630 $\pm$ 0.010	0.622 $\pm$ 0.033	0.613 $\pm$ 0.023	0.582 $\pm$ 0.025	0.572 $\pm$ 0.022	0.604
PathQ-Former w/o aux heads	WSI+RNA	0.609 $\pm$ 0.016	0.611 $\pm$ 0.039	0.638 $\pm$ 0.031	0.575 $\pm$ 0.012	0.593 $\pm$ 0.015	0.605
SurvPath	WSI+RNA	0.594 $\pm$ 0.012	0.536 $\pm$ 0.026	0.570 $\pm$ 0.038	0.552 $\pm$ 0.019	0.587 $\pm$ 0.008	0.568
Late fusion	WSI+RNA	0.629 $\pm$ 0.018	0.606 $\pm$ 0.020	0.620 $\pm$ 0.069	0.571 $\pm$ 0.009	0.544 $\pm$ 0.006	0.594
ABMIL	WSI	0.566 $\pm$ 0.011	0.575 $\pm$ 0.004	0.587 $\pm$ 0.023	0.562 $\pm$ 0.028	0.553 $\pm$ 0.020	0.569
Slide-only PathQ-Former	WSI	0.592 $\pm$ 0.003	0.615 $\pm$ 0.027	0.574 $\pm$ 0.024	0.585 $\pm$ 0.003	0.582 $\pm$ 0.019	0.590
SNN	RNA	0.590 $\pm$ 0.009	0.558 $\pm$ 0.013	0.572 $\pm$ 0.028	0.535 $\pm$ 0.006	0.549 $\pm$ 0.009	0.561
RNA MLP	RNA	0.611 $\pm$ 0.012	0.627 $\pm$ 0.013	0.633 $\pm$ 0.016	0.555 $\pm$ 0.012	0.532 $\pm$ 0.009	0.592
RNA-only PathQ-Former	RNA	0.608 $\pm$ 0.020	0.559 $\pm$ 0.001	0.598 $\pm$ 0.035	0.554 $\pm$ 0.013	0.511 $\pm$ 0.005	0.566

Table 6: Seed-averaged paired tests over the 25 pooled (cohort, fold) units: mean difference in C-index (first minus second method), 95% percentile-bootstrap CI (10,000 resamples of folds), paired t -test and exact Wilcoxon p -values, and folds won. Seeds are averaged per fold before pairing (three seeds for DSS; seeds 0–1 against late fusion; one seed for OS).

Comparison	Setting	Δ	95% CI	t p	W p	Wins
PathQ-Former vs SurvPath	DSS	+0.036	[−0.004, +0.071]	0.074	0.030	18/25
PathQ-Former w/o aux heads vs SurvPath	DSS	+0.037	[+0.009, +0.065]	0.020	0.020	17/25
ABMIL vs SurvPath	DSS	+0.001	[−0.024, +0.026]	0.95	0.85	14/25
SNN vs SurvPath	DSS	−0.007	[−0.060, +0.045]	0.80	0.73	10/25
RNA MLP vs SurvPath	DSS	+0.024	[−0.027, +0.077]	0.38	0.41	15/25
PathQ-Former vs RNA MLP	DSS	+0.012	[−0.022, +0.044]	0.47	0.34	14/25
PathQ-Former w/o aux heads vs RNA MLP	DSS	+0.014	[−0.023, +0.049]	0.47	0.37	16/25
PathQ-Former vs late fusion	DSS, seeds 0–1	+0.011	[−0.013, +0.034]	0.39	0.28	14/25
PathQ-Former w/o aux heads vs late fusion	DSS, seeds 0–1	+0.009	[−0.020, +0.036]	0.54	0.52	14/25
PathQ-Former vs SurvPath	OS, seed 0	+0.034	[+0.014, +0.054]	0.003	0.003	19/25
PathQ-Former w/o aux heads vs SurvPath	DSS, 10 epochs	+0.028	[−0.001, +0.057]	0.079	0.063	18/25

Against the RNA-only MLP the margins are +0.012 and +0.014 ($p \geq 0.34$), and against the late-fusion ensemble +0.011 and +0.009 ($p \geq 0.28$; seeds 0–1); the MLP scores higher than PathQ-Former on BRCA and COADREAD and higher than the variant without auxiliary heads on BLCA and BRCA. No unimodal baseline differs from SurvPath pooled (ABMIL +0.001, SNN −0.007, MLP +0.024). The equal-budget result is therefore parity with the best unimodal baseline, not superiority. On overall survival (one seed) PathQ-Former leads SurvPath by 0.034 ([+0.014, +0.054]; $p = 0.003$ and 0.003), and under a 10-epoch budget the variant without auxiliary heads leads by 0.028 ($p = 0.079$ and 0.063); both are tabulated in Tables 10 and 11.

4.3 Missing inputs at test time

Figure 3 shows the random-missingness sweep. With half of the validation patients missing one modality, the fused C-index of PathQ-Former changes by at most 0.025 (BRCA, 0.622 to 0.597) and that of the variant without auxiliary heads by at most 0.018 (BRCA). For PathQ-Former every change is within one sample seed standard deviation of the complete-input score (Table 5); for the variant without auxiliary heads the HNSC and STAD changes (0.013 and 0.017) exceed it by 0.001 and 0.002. A site with mixed availability can instead route slide-only patients to ABMIL and RNA-only patients to the RNA MLP, or average the slide-only and RNA-only models when both inputs exist; routing and late fusion are robust by construction, and the late-fusion ensemble matches the joint model (Table 6). The one-checkpoint result is about coverage, not accuracy: with the RNA removed, the single PathQ-Former checkpoint scores 0.576–0.600 against 0.553–0.587 for ABMIL (higher on four cohorts, lower on COADREAD); with the slide removed it scores 0.538–0.617 against 0.532–0.633 for the RNA MLP, higher on BLCA and STAD and lower on BRCA, COADREAD and HNSC (by up to 0.050 on COADREAD; three-seed means throughout).

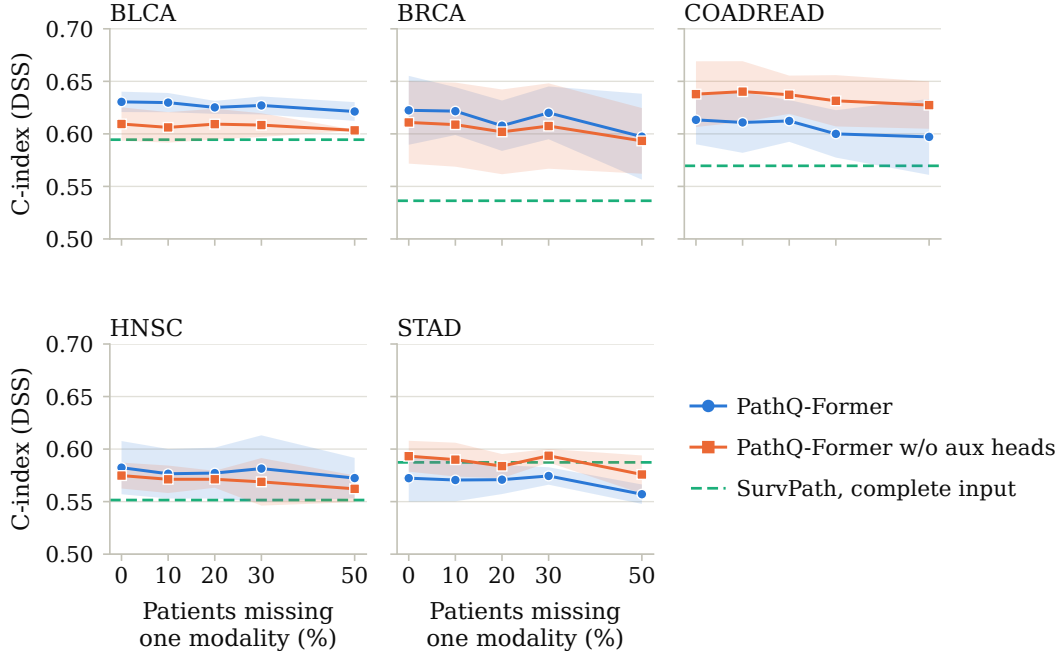


Figure 3: C-index of PathQ-Former and of its variant without auxiliary heads (20 epochs) when 0–50% of validation patients are missing one modality at random (mean of five random draws per fold). Lines: mean over three seeds of the five-fold mean; bands: ± 1 sd over seeds. Dashed line: SurvPath with complete inputs (three-seed mean). With half the patients missing a modality the fused C-index changes by at most 0.025 (BRCA).

4.4 Where the robustness comes from

Evaluating the same checkpoints under mean imputation instead of the null code separates the two. For the variant without auxiliary heads, replacing the removed slide by the training-mean patch embedding instead of the null code turns a loss of 0.030 into a loss of 0.069 (t -test $p = 3 \times 10^{-4}$, Wilcoxon $p = 2.5 \times 10^{-4}$; one seed, 25 folds), the size of SurvPath’s collapse; with the heads the loss is 0.035 under mean imputation ($p = 0.040$ and 0.045 ; three seeds) against 0.025 with the null code (per-cohort values in Table 14). The attribution is therefore: null codes and modality dropout remove the collapse without the slide, and the null code is what makes the checkpoint usable with the slide absent; the auxiliary heads are what makes the fused output respond to RNA; and the fused risk remains histology-led (ρ 0.83 against 0.45). Fusion-block ablations on BLCA under the 10-epoch budget (modality dropout 0–0.5, 16–64 queries, fusion depth 1–3; one seed) move the fused C-index within 0.595–0.638, inside the fold standard deviation of 0.07–0.13 (Table 15).

4.5 Checkpoint selection on event-poor folds

Every run stores its per-epoch validation history, so the effect of a selection rule can be replayed without training. Table 7 replays validation-loss early stopping with patience 5 on the 590 stored fold histories of the eight 20-epoch methods. The rule selects epoch 1 on median (mean 1.6; 93% of folds epoch 1–3) and would have stopped after 6.6 of 20 epochs on average, with the patience limit reached in every fold. Pooled over the 25 seed-averaged folds, the early-stopped checkpoint scores below the final one by 0.090 for PathQ-Former without heads, 0.059 for PathQ-Former, 0.065 for the slide-only and 0.034 for the RNA-only PathQ-Former, 0.038 for SNN and 0.027 for the RNA MLP, and by nothing for ABMIL (−0.001) and SurvPath (+0.000). The histories show why: ABMIL and SurvPath reach their best validation C-index at epoch 6.2 and 6.9 on average, SNN and the MLP at 7.2–7.3, and the PathQ-Former variants, which train with a warm-up epoch and a lower learning rate (Table 8) and are the late learners, at 9.3–10.0, so an epoch-1–2 checkpoint costs the late learners most. Under the replayed rule 23 of the 40 method $\times$ cohort cells have a mean C-index

Table 7: Replay of validation-loss early stopping (patience 5) on the stored 20-epoch histories (three seeds; the single-modality PathQ-Formers 70 instead of 75 fold histories). Final and early-stopped C-index: pooled mean over (fold, seed) runs. Δ : early-stopped minus final, paired over the 25 seed-averaged folds, with t -test and Wilcoxon p -values. Selected: mean epoch the rule selects; best val.: mean epoch of the highest validation C-index. Selection and scoring use the same fold, so Δ is biased in favour of the rule. For the two single-modality PathQ-Formers this run-weighted mean differs by 0.001–0.002 from the Pooled column of Table 5, which averages the five cohort means.

Method	Final	Early-stopped	Δ	$t\ p$	W p	Selected	Best val.
PathQ-Former	0.604	0.546	−0.059	0.016	8×10^{-4}	2.2	10.0
PathQ-Former w/o aux heads	0.605	0.516	−0.090	3×10^{-7}	1×10^{-6}	1.8	9.9
Slide-only PathQ-Former	0.588	0.523	−0.065	0.006	0.005	1.8	9.3
RNA-only PathQ-Former	0.567	0.532	−0.034	0.062	0.080	1.4	9.7
SurvPath (official)	0.568	0.568	+0.000	0.99	0.98	1.6	6.9
ABMIL	0.569	0.568	−0.001	0.95	0.61	1.4	6.2
SNN	0.561	0.523	−0.038	0.054	0.052	1.4	7.3
RNA MLP	0.592	0.565	−0.027	0.078	0.075	1.2	7.2

within 0.45–0.55, against 5 under the final checkpoint (HNSC 8 of 8, STAD 7 of 8, BRCA 4 of 8). An earlier validation-loss run on these cohorts (120 folds) behaved the same way: median selected epoch 1 and 13 of 22 method $\times$ cohort cells within 0.45–0.55 (Appendix G). Every replayed rule selects and scores on the same fold, so the measured deficit of early stopping is conservative. On cohorts with 5–45 events per validation fold, checkpoints should not be selected on the validation negative log-likelihood; a fixed budget with the final checkpoint, or at least the selected epochs and the events per fold, should be reported.

5 Discussion

A general lesson Fusion of a strong and a weak modality can converge to ignoring the weak one with no accuracy signal: SurvPath’s fused C-index is as high as its slide-only behaviour permits, and nothing in the survival objective penalises leaving the genomic branch unused once the fusion block has found a low-loss route through the histology branch. This is the modality-competition effect that Gradient Blending (Wang et al., 2020) and OGM-GE (Peng et al., 2022) describe for classification, and the same remedies apply: per-modality supervision supplies a direct gradient to the weak branch. The audit detects the failure without any training: removal and permutation across patients, with the risk-score correlation, show which input a ranking responds to, and they should accompany any multimodal C-index. The audit also shows what unimodal supervision does and does not do here. It makes the fused output respond to RNA, but the fused risk still correlates at 0.83 with the same checkpoint’s RNA-removed risk and at 0.45 with its slide-removed risk, and the RNA-permuted loss (0.023) is a third of the slide-permuted loss (0.067).

The RNA-only baseline A two-layer MLP on bulk expression, run with repository defaults, matches both fusion variants pooled and scores higher than PathQ-Former on two of five cohorts. This baseline is cheap and seldom reported at equal budget; without it, an improvement attributed to fusion may be an improvement attributable to the genomic branch. Separating “pathway tokens help” from “fusion helps” would need an MLP that receives the same pathway tokens, which we have not run.

Limitations

- **Baseline tuning and variant selection.** SurvPath was not re-tuned, and PathQ-Former’s hyper-parameters and the auxiliary-head variant were chosen as described under Scope of claims and in Appendix A. The accuracy comparison is favourable to PathQ-Former by an amount that cannot be bounded without tuning the baseline.
- **Recipe on other architectures.** Null codes, modality dropout and auxiliary heads have not been applied to SurvPath, so how much of the robustness is specific to the query bottleneck is untested.

- **Seeds and statistics.** Three seeds for the main comparison, two for late fusion, one for OS. Per-cohort tests have five folds; the fold-level standard deviation of the C-index is 0.12–0.25 on COADREAD for the fusion models (0.05–0.25 over all methods), fold 3 of BLCA is weak for every method except the RNA-only PathQ-Former, and fold 0 of COADREAD, with five events, scores 0.85–0.92 for PathQ-Former across seeds. No per-cohort DSS difference survives correction; on OS (one seed) BLCA, the development cohort, does under the t -test only (Holm $p = 0.039$; Wilcoxon 0.312). The RNA-permuted loss of 0.023 that supports the auxiliary-head claim has $p = 0.075$ and 0.063 over 25 folds.
- **Data.** TCGA only, with frozen UNI2-h features whose pretraining corpus may include TCGA slides, which would make every WSI-based result optimistic; no external cohort; missingness simulated at random, so informative missingness is not covered; no clinical covariates (stage, grade, age), so whether any model adds to them is untested.

6 Conclusion

A fused histology–transcriptomics survival model can score well while its ranking ignores one input: the official SurvPath implementation, under an equal-budget protocol on five TCGA cohorts, does not respond to its RNA input and collapses without the slide. Removal and permutation audits with risk-score correlations detect this and should be reported with any multimodal C-index. Training one fusion model with null codes and modality dropout removes the collapse, and auxiliary unimodal heads make the fused prediction respond to RNA; the resulting single checkpoint serves complete, slide-only and RNA-only patients; with both inputs it is not detectably different from an RNA-only MLP or a late-fusion ensemble, and with one input absent it stays within 0.05 of the dedicated unimodal baseline. Replaying checkpoint selection on the stored histories shows that validation-loss early stopping would have selected epoch 1–3 checkpoints and penalised the late-learning models on these event-poor folds, which argues for fixed budgets and for reporting selected epochs and events per fold.

Data and Code Availability

TCGA data are public and de-identified. UNI2-h patch features (Mahmood Lab, 2025a;b) and the RNA-seq matrices, clinical metadata and splits released with SurvPath (Jaume et al., 2024) are available from their authors. The code (MIT licence), configuration files, archived per-run results (results, configuration, per-fold predictions and training histories for every run; inventory in Appendix G) and the script that regenerates every table in Section 4 and Appendix B are included in the anonymized supplementary material. The 275 trained fold models (PathQ-Former and its variant without auxiliary heads at three seeds, SurvPath and the slide-only and RNA-only PathQ-Formers at seed 0, on the five cohorts, plus the two OS runs: 11 run configurations $\times$ five cohorts $\times$ five folds; 18 GB) will be released with the camera-ready version.

Broader Impact Statement

The model is a research prototype evaluated on public, de-identified TCGA data with simulated missingness. The model has not been validated on an external cohort, compared with clinical prognostic factors, or tested under informative missingness, and it must not be used to inform patient care. Performance across sex, race and age subgroups was not evaluated. The evaluation findings, that a fusion model can score well while its ranking is invariant to one input and that validation-loss checkpoint selection can leave methods near chance on event-poor cohorts, argue for reporting input-use audits, selection rules and events per fold in this literature.

References

Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi,

- Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. In *Advances in Neural Information Processing Systems*, volume 35, pp. 23716–23736, 2022.
- Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*, volume 12346 of *Lecture Notes in Computer Science*, pp. 213–229, 2020.
- Anika Cheerla and Olivier Gevaert. Deep learning with multimodal representation for pancancer prognosis prediction. *Bioinformatics*, 35(14):i446–i454, 2019.
- Richard J. Chen, Ming Y. Lu, Wei-Hung Weng, Tiffany Y. Chen, Drew F. K. Williamson, Trevor Manz, Maha Shady, and Faisal Mahmood. Multimodal co-attention transformer for survival prediction in gigapixel whole slide images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4015–4025, 2021.
- Richard J. Chen, Ming Y. Lu, Jingwen Wang, Drew F. K. Williamson, Scott J. Rodig, Neal I. Lindeman, and Faisal Mahmood. Pathomic fusion: An integrated framework for fusing histopathology and genomic features for cancer diagnosis and prognosis. *IEEE Transactions on Medical Imaging*, 41(4):757–770, 2022a.
- Richard J. Chen, Ming Y. Lu, Drew F. K. Williamson, Tiffany Y. Chen, Jana Lipkova, Zahra Noor, Muhammad Shaban, Maha Shady, Mane Williams, Bumjin Joo, and Faisal Mahmood. Pan-cancer integrative histology-genomic analysis via multimodal deep learning. *Cancer Cell*, 40(8):865–878.e6, 2022b.
- Richard J. Chen, Tong Ding, Ming Y. Lu, Drew F. K. Williamson, Guillaume Jaume, Andrew H. Song, Bowen Chen, Andrew Zhang, Daniel Shao, Muhammad Shaban, Mane Williams, Lukas Oldenburg, Luca L. Weishaupt, Judy J. Wang, Anurag Vaidya, Long Phi Le, Georg Gerber, Sharifa Sahai, Walt Williams, and Faisal Mahmood. Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30:850–862, 2024.
- Marc Gillespie, Bijay Jassal, Ralf Stephan, Marija Milacic, Karen Rothfels, Andrea Senff-Ribeiro, Johannes Griss, Cristoffer Sevilla, Lisa Matthews, Chuqiao Gong, Chuan Deng, Thawfeek Varusai, Eliot Ragueneau, Yusra Haider, Bruce May, Veronica Shamovsky, Joel Weiser, Timothy Brunson, Nasim Sanati, Liam Beckman, Xiang Shao, Antonio Fabregat, Konstantinos Sidiropoulos, Julieth Murillo, Guilherme Viteri, Justin Cook, Solomon Shorser, Gary Bader, Emek Demir, Chris Sander, Robin Haw, Guanming Wu, Lincoln Stein, Henning Hermjakob, and Peter D’Eustachio. The reactome pathway knowledgebase 2022. *Nucleic Acids Research*, 50(D1):D687–D692, 2022.
- Erika Graf, Claudia Schmoor, Willi Sauerbrei, and Martin Schumacher. Assessment and comparison of prognostic classification schemes for survival data. *Statistics in Medicine*, 18(17–18):2529–2545, 1999.
- Frank E. Harrell, Robert M. Califf, David B. Pryor, Kerry L. Lee, and Robert A. Rosati. Evaluating the yield of medical tests. *JAMA*, 247(18):2543–2546, 1982.
- Maximilian Ilse, Jakub Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *Proceedings of Machine Learning Research*, pp. 2127–2136, 2018.
- Andrew Jaegle, Felix Gimeno, Andy Brock, Oriol Vinyals, Andrew Zisserman, and João Carreira. Perceiver: General perception with iterative attention. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pp. 4651–4664, 2021.
- Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 3543–3556, 2019.

- Guillaume Jaume, Anurag Vaidya, Richard J. Chen, Drew F. K. Williamson, Paul Pu Liang, and Faisal Mahmood. Modeling dense multimodal interactions between biological pathways and histology for survival prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11579–11590, 2024.
- Günter Klambauer, Thomas Unterthiner, Andreas Mayr, and Sepp Hochreiter. Self-normalizing neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Juho Lee, Yoonho Lee, Jungtaek Kim, Adam Kosiorek, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pp. 3744–3753, 2019.
- Yi-Lun Lee, Yi-Hsuan Tsai, Wei-Chen Chiu, and Chen-Yu Lee. Multimodal prompting with missing modalities for visual recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14943–14952, 2023.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pp. 19730–19742, 2023.
- Arthur Liberzon, Chet Birger, Helga Thorvaldsdóttir, Mahmoud Ghandi, Jill P. Mesirov, and Pablo Tamayo. The molecular signatures database (MSigDB) hallmark gene set collection. *Cell Systems*, 1(6):417–425, 2015.
- Fengchun Liu, Linghan Cai, Zhikang Wang, Zhiyuan Fan, Jin-gang Yu, Hao Chen, and Yongbing Zhang. Prototype-guided cross-modal knowledge enhancement for adaptive survival prediction. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, volume 15965 of *Lecture Notes in Computer Science*, pp. 522–532, 2025a. doi: 10.1007/978-3-032-04978-0_50.
- Hong Liu, Haosen Yang, Federica Eduati, Josien P. W. Pluim, and Mitko Veta. Adaptive prototype learning for multimodal cancer survival analysis. *arXiv preprint arXiv:2503.04643*, 2025b.
- Jianfang Liu, Tara Lichtenberg, Katherine A. Hoadley, Laila M. Poisson, Alexander J. Lazar, Andrew D. Cherniack, Albert J. Kovatich, Christopher C. Benz, Douglas A. Levine, Adrian V. Lee, Larsson Omberg, Denise M. Wolf, Craig D. Shriver, Vesteynn Thorsson, The Cancer Genome Atlas Research Network, and Hai Hu. An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics. *Cell*, 173(2):400–416.e11, 2018.
- Liyuan Liu, Haoming Jiang, Pengcheng He, Weizhu Chen, Xiaodong Liu, Jianfeng Gao, and Jiawei Han. On the variance of the adaptive learning rate and beyond. In *International Conference on Learning Representations (ICLR)*, 2020.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- Ming Y. Lu, Drew F. K. Williamson, Tiffany Y. Chen, Richard J. Chen, Matteo Barbieri, and Faisal Mahmood. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering*, 5(6):555–570, 2021.
- Mengmeng Ma, Jian Ren, Long Zhao, Sergey Tulyakov, Cathy Wu, and Xi Peng. SMIL: Multimodal learning with severely missing modality. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pp. 2302–2310, 2021.
- Mengmeng Ma, Jian Ren, Long Zhao, Davide Testuggine, and Xi Peng. Are multimodal transformers robust to missing modality? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 18177–18186, 2022.

- Mahmood Lab. UNI2-h: pathology foundation model and precomputed TCGA patch features. Hugging Face model repository, <https://huggingface.co/MahmoodLab/UNI2-h>, 2025a. Precomputed patch features are distributed separately as the Hugging Face dataset <https://huggingface.co/datasets/MahmoodLab/UNI2-h-features>. Accessed September 2026.
- Mahmood Lab. UNI2-h-features: precomputed UNI2-h patch features for TCGA, CPTAC and PANDA whole-slide images. Hugging Face dataset repository, <https://huggingface.co/datasets/MahmoodLab/UNI2-h-features>, 2025b. 256×256 patches at 20× magnification, HDF5 coordinates and features; licence CC-BY-NC-ND 4.0. Accessed September 2026.
- Pooya Mobadersany, Safoora Yousefi, Mohamed Amgad, David A. Gutman, Jill S. Barnholtz-Sloan, José E. Velázquez Vega, Daniel J. Brat, and Lee A. D. Cooper. Predicting cancer outcomes from histology and genomics using convolutional networks. *Proceedings of the National Academy of Sciences*, 115(13):E2970–E2979, 2018.
- Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003. doi: 10.1023/A:1024068626366.
- Natalia Neverova, Christian Wolf, Graham Taylor, and Florian Nebout. ModDrop: Adaptive multi-modal gesture recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 38(8):1692–1706, 2016.
- Jie Peng, Shuang Zhou, Longwei Yang, Yiran Song, Mohan Zhang, Kaixiong Zhou, Feng Xie, Mingquan Lin, Rui Zhang, and Tianlong Chen. Continually evolved multimodal foundation models for cancer prognosis. *arXiv preprint arXiv:2501.18170*, 2025.
- Xiaokang Peng, Yake Wei, Andong Deng, Dong Wang, and Di Hu. Balanced multimodal learning via on-the-fly gradient modulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 8238–8247, 2022. doi: 10.1109/CVPR52688.2022.00806.
- Sebastian Pölsterl. scikit-survival: A library for time-to-event analysis built on top of scikit-learn. *Journal of Machine Learning Research*, 21(212):1–6, 2020.
- Linhao Qu, Dan Huang, Shaoting Zhang, and Xiaosong Wang. Multi-modal data binding for survival analysis modeling with incomplete data and annotations. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, volume 15005 of *Lecture Notes in Computer Science*, pp. 501–510, 2024. doi: 10.1007/978-3-031-72086-4_47.
- Mingcheng Qu, Guang Yang, Donglin Di, Yue Gao, Tonghua Su, Yang Song, and Lei Fan. Memory-augmented incomplete multimodal survival prediction via cross-slide and gene-attentive hypergraph learning. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, volume 15969 of *Lecture Notes in Computer Science*, pp. 318–327, 2025. doi: 10.1007/978-3-032-05127-1_31.
- Zhuchen Shao, Hao Bian, Yang Chen, Yifeng Wang, Jian Zhang, Xiangyang Ji, and Yongbing Zhang. TransMIL: Transformer based correlated multiple instance learning for whole slide image classification. In *Advances in Neural Information Processing Systems*, volume 34, pp. 2136–2147, 2021.
- Andrew H. Song, Richard J. Chen, Guillaume Jaume, Anurag J. Vaidya, Alexander S. Baras, and Faisal Mahmood. Multimodal prototyping for cancer survival prediction. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 46050–46073, 2024.
- Hajime Uno, Tianxi Cai, Michael J. Pencina, Ralph B. D’Agostino, and L. J. Wei. On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. *Statistics in Medicine*, 30(10):1105–1117, 2011.
- Luís A. Vale-Silva and Karl Rohr. Long-term cancer survival prediction using multimodal deep learning. *Scientific Reports*, 11:13505, 2021.

- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, Eric Zimmermann, James Hall, Neil Tenenholtz, Nicolo Fusi, Ellen Yang, Philippe Mathieu, Alexander van Eck, Donghun Lee, Julian Viret, Eric Robert, Yi Kan Wang, Jeremy D. Kunz, Matthew C. H. Lee, Jan H. Bernhard, Ran A. Godrich, Gerard Oakley, Ewan Millar, Matthew Hanna, Hannah Wen, Juan A. Retamero, William A. Moye, Razik Yousfi, Christopher Kanan, David S. Klimstra, Brandon Rothrock, Siqi Liu, and Thomas J. Fuchs. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30:2924–2935, 2024.
- Hu Wang, Yuanhong Chen, Congbo Ma, Jodie Avery, Louise Hull, and Gustavo Carneiro. Multi-modal learning with missing modality via shared-specific feature modelling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 15878–15887, 2023.
- Weiyao Wang, Du Tran, and Matt Feiszli. What makes training multi-modal classification networks hard? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 12692–12702, 2020. doi: 10.1109/CVPR42600.2020.01271.
- Xiyue Wang, Sen Yang, Jun Zhang, Minghui Wang, Jing Zhang, Wei Yang, Junzhou Huang, and Xiao Han. Transformer-based unsupervised contrastive learning for histopathological image classification. *Medical Image Analysis*, 81:102559, 2022.
- Zhikang Wang, Yumeng Zhang, Yingxue Xu, Seiya Imoto, Hao Chen, and Jiangning Song. Histo-genomic knowledge association for cancer prognosis from histopathology whole slide images. *IEEE Transactions on Medical Imaging*, 44(5):2170–2181, 2025. doi: 10.1109/TMI.2025.3526816.
- Hanwen Xu, Naoto Usuyama, Jaspreet Bagga, Sheng Zhang, Rajesh Rao, Tristan Naumann, Cliff Wong, Zelalem Gero, Javier González, Yu Gu, Yanbo Xu, Mu Wei, Wenhui Wang, Shuming Ma, Furu Wei, Jianwei Yang, Chunyuan Li, Jianfeng Gao, Jaylen Rosemon, Tucker Bower, Soohee Lee, Roshanthi Weerasinghe, Bill J. Wright, Ari Robicsek, Brian Piening, Carlo Bifulco, Sheng Wang, and Hoifung Poon. A whole-slide foundation model for digital pathology from real-world data. *Nature*, 630:181–188, 2024.
- Yingxue Xu and Hao Chen. Multimodal optimal transport-based co-attention transformer with global structure consistency for survival prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 21241–21251, 2023.
- Yingxue Xu, Fengtao Zhou, Chenyu Zhao, Yihui Wang, Can Yang, and Hao Chen. Distilled prompt learning for incomplete multimodal survival prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5102–5111, 2025. doi: 10.1109/CVPR52734.2025.00481.
- Kai Yu, Shuang Zhou, Yiran Song, Zaifu Zhan, Jie Peng, Kaixiong Zhou, Tianlong Chen, Feng Xie, Meng Wang, Huazhu Fu, Mingquan Lin, and Rui Zhang. PRIME: Prototype-driven multimodal pretraining for cancer prognosis with missing modalities. *arXiv preprint arXiv:2604.04999*, 2026.
- Shekoufeh Gorgi Zadeh and Matthias Schmid. Bias in cross-entropy-based training of deep survival networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(9):3126–3137, 2021.
- Yilan Zhang, Yingxue Xu, Jianqi Chen, Fengying Xie, and Hao Chen. Prototypical information bottlenecking and disentangling for multimodal cancer survival prediction. In *International Conference on Learning Representations (ICLR)*, 2024.
- Fengtao Zhou and Hao Chen. Cross-modal translation and alignment for survival analysis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 21485–21494, 2023.
- Junjie Zhou, Jiao Tang, Yingli Zuo, Peng Wan, Daoqiang Zhang, and Wei Shao. Robust multimodal survival prediction with conditional latent differentiation variational autoencoder. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10384–10393, 2025. doi: 10.1109/CVPR52734.2025.00971.

A Training Details and Selection History

Query blocks Each modality $m \in \{h, g\}$ has $K = 32$ learned queries $Q^m \in \mathbb{R}^{K \times d}$ with $d = 256$ and a stack of $L = 2$ pre-norm blocks,

$$Q \leftarrow Q + \text{CA}(\text{LN}(Q), U^m), \quad (2)$$

$$Q \leftarrow Q + \text{SA}(\text{LN}(Q)), \quad (3)$$

$$Q \leftarrow Q + \text{FFN}(\text{LN}(Q)), \quad (4)$$

where U^m are the modality’s input tokens (the LayerNormed patch embeddings, or the 275 pathway tokens produced by per-pathway two-layer MLPs of hidden width 128), CA is 8-head cross-attention with queries from Q and keys and values from U^m , SA is self-attention among the queries and FFN is a two-layer feed-forward network. Cross-attention over N input tokens costs $O(KNd)$ and attention among the queries $O(K^2d)$; the block returns 32 vectors, so the fusion stage does not depend on N . The fused codes pass through two Transformer encoder layers ($d = 256$, 8 heads) and a learned-query attention pooling; a linear layer with a sigmoid produces the $T = 4$ hazards. Table 8 gives the training configuration.

Table 8: Training configuration of PathQ-Former and of the official SurvPath baseline.

Setting	PathQ-Former	SurvPath (official code)
Epochs / selection	20, final checkpoint	20, final checkpoint
Optimizer	AdamW (Loshchilov & Hutter, 2019), lr 10^{-4} , wd 10^{-4} , 1 warm-up epoch, cosine decay	RAdam (Liu et al., 2020), lr 5×10^{-4} (published), wd 10^{-4}
Batch	1 patient, gradient accumulation 2, gradient clip 1.0	1 patient
Dropout	0.25; modality dropout 0.15	0.1
Patches	all patches of all slides	4,096-patch subsample (published)
Loss	Eq. (1), censored term unweighted; auxiliary heads $\lambda = 0.5$ (full model) or 0	Eq. (1), censored term weighted 0.5 (published)
Sampling	uniform over training patients	weighted by (bin, censorship) (published)
Genes	min-max to $[-1, 1]$, fit on the training split	same
Parameters	16.9 M	21.2 M

Selection history The base hyper-parameters of PathQ-Former (learning-rate schedule, dropout, modality-dropout rate, number of queries, fusion depth) were chosen on BLCA under the 10-epoch budget from the single-seed ablation runs in Table 15; none of those settings is distinguishable from the others at one seed, and the chosen values are defaults rather than optima. The budget was raised from 10 to 20 epochs after inspecting the pooled validation curves of PathQ-Former and SurvPath under the 10-epoch budget; the 20-epoch curves (Figure 4) show SurvPath flat from about epoch 7 and both PathQ-Former variants rising until about epoch 13–18. The learning rate and gradient-accumulation setting of PathQ-Former were changed at the same time as the budget, so the 10- and 20-epoch results are not a pure budget comparison; a single-seed check at lr 2×10^{-4} on BLCA did not improve on the default (0.615 vs 0.630). The auxiliary-head variant and $\lambda = 0.5$ were adopted after the 20-epoch results of the variant without auxiliary heads on all five cohorts had been seen; the two variants are therefore reported as co-primary throughout, and no per-cohort result on BLCA is treated as confirmatory. Best-epoch validation scores were logged in every run and are used only in the selection replay of Section 4.5.

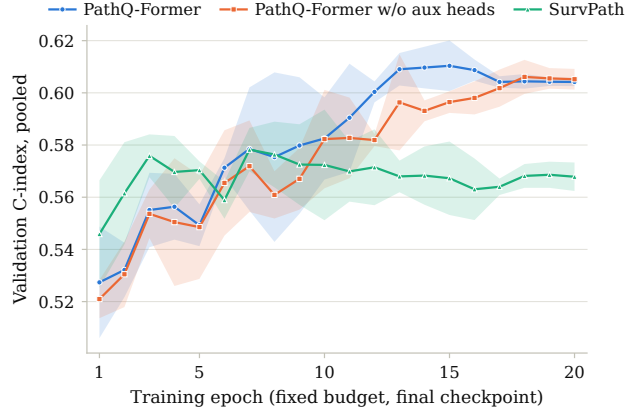


Figure 4: Validation C-index per training epoch under the 20-epoch budget, pooled over all cohorts and folds; mean over three seeds with a ± 1 sd band over seeds, for PathQ-Former, its variant without auxiliary heads, and SurvPath. All results in the paper use the final epoch; the curves were inspected only to set the budget.

B Full Tables

Table 9: Per-cohort seed-averaged paired differences in C-index against SurvPath (five folds per cohort; three seeds for DSS, one seed for OS). Raw and Holm-adjusted p -values (over the five cohorts, t and Wilcoxon adjusted separately) are computed from unrounded values. [†]Development cohort.

	BLCA [†]	BRCA	COADREAD	HNSC	STAD
<i>PathQ-Former vs SurvPath (DSS, three seeds)</i>					
Δ	+0.036	+0.086	+0.044	+0.031	−0.015
t p (Holm)	0.175 (0.700)	0.017 (0.084)	0.583 (1.000)	0.544 (1.000)	0.719 (1.000)
W p (Holm)	0.312 (1.000)	0.062 (0.312)	0.625 (1.000)	0.625 (1.000)	0.625 (1.000)
wins	3/5	5/5	4/5	4/5	2/5
<i>PathQ-Former w/o aux heads vs SurvPath (DSS, three seeds)</i>					
Δ	+0.015	+0.075	+0.068	+0.023	+0.006
t p (Holm)	0.518 (1.000)	0.073 (0.367)	0.202 (0.807)	0.580 (1.000)	0.847 (1.000)
W p (Holm)	0.625 (1.000)	0.125 (0.625)	0.188 (0.750)	0.625 (1.000)	0.812 (1.000)
wins	3/5	4/5	4/5	4/5	2/5
<i>PathQ-Former vs SurvPath (OS, seed 0)</i>					
Δ	+0.043	+0.073	−0.015	+0.025	+0.044
t p (Holm)	0.008 (0.039)	0.015 (0.058)	0.578 (0.578)	0.046 (0.138)	0.230 (0.460)
W p (Holm)	0.062 (0.312)	0.062 (0.312)	0.625 (0.625)	0.125 (0.375)	0.312 (0.625)
wins	5/5	5/5	2/5	4/5	3/5

Table 12 reports the secondary metrics. Uno’s C-index tracks Harrell’s: PathQ-Former leads SurvPath by 0.039 ($[-0.016, +0.086]$; Wilcoxon $p = 0.042$) and the variant without auxiliary heads by 0.046 ($[+0.004, +0.087]$; $p = 0.045$ and 0.042). Each of the four other methods in Table 12 beats SurvPath on the integrated Brier score by about 0.07 (Wilcoxon $p < 10^{-4}$): SurvPath’s hazards are poorly calibrated under this protocol (its published loss weights the censored term by 0.5), so the IBS is not a fair accuracy comparison and is reported for completeness only.

Table 10: Overall survival (20 epochs, seed 0): five-fold mean C-index $\pm$ standard deviation over the five folds, and the PathQ-Former checkpoint evaluated with one input removed (null codes). Pooled: mean over the five cohorts. [†]Development cohort.

Cohort	PathQ-Former	SurvPath	Δ	PathQ-Former, one input removed	
				RNA removed	Slide removed
BLCA [†]	0.595 $\pm$ 0.053	0.552 $\pm$ 0.056	+0.043	0.576	0.592
BRCA	0.596 $\pm$ 0.043	0.523 $\pm$ 0.052	+0.073	0.531	0.556
COADREAD	0.581 $\pm$ 0.132	0.596 $\pm$ 0.153	-0.015	0.586	0.562
HNSC	0.536 $\pm$ 0.036	0.510 $\pm$ 0.029	+0.025	0.535	0.550
STAD	0.603 $\pm$ 0.070	0.559 $\pm$ 0.045	+0.044	0.574	0.582
Pooled (25 folds)	0.582	0.548	+0.034		

Table 11: 10-epoch budget (DSS, three seeds): C-index mean $\pm$ sample standard deviation over seeds of the five-fold mean for PathQ-Former without auxiliary heads and SurvPath, with the seed-averaged paired difference over the five folds (t -test and Wilcoxon p -values). Pooled: 25 folds, 18 of 25 won. [†]Development cohort.

Cohort	PathQ-Former w/o aux heads	SurvPath	Δ	t p	W p
BLCA [†]	0.618 $\pm$ 0.014	0.589 $\pm$ 0.004	+0.029	0.272	0.312
BRCA	0.600 $\pm$ 0.031	0.536 $\pm$ 0.015	+0.064	0.142	0.188
COADREAD	0.646 $\pm$ 0.046	0.551 $\pm$ 0.022	+0.094	0.015	0.062
HNSC	0.556 $\pm$ 0.007	0.568 $\pm$ 0.035	-0.013	0.595	1.000
STAD	0.574 $\pm$ 0.025	0.607 $\pm$ 0.012	-0.033	0.471	0.625
Pooled	0.599	0.570	+0.028	0.079	0.063

Table 12: Secondary metrics under the 20-epoch DSS protocol: mean $\pm$ sample standard deviation over three seeds of the five-fold mean. Uno’s IPCW C-index is truncated at the 75th percentile of training event times; the integrated Brier score (IBS, lower is better) is evaluated on the interior quartile grid and is undefined on BRCA fold 4 for every method (cells marked * average four folds; the fold is dropped from the paired units, $n = 24$). Δ vs SurvPath: seed-averaged paired difference over the pooled folds with 95% bootstrap CI, t -test and Wilcoxon p -values.

	PathQ-Former	PathQ-Former w/o aux heads	ABMIL	RNA MLP	SurvPath
<i>Uno’s IPCW C-index</i>					
BLCA	0.630 $\pm$ 0.009	0.612 $\pm$ 0.019	0.560 $\pm$ 0.014	0.605 $\pm$ 0.013	0.597 $\pm$ 0.010
BRCA	0.636 $\pm$ 0.012	0.648 $\pm$ 0.038	0.599 $\pm$ 0.012	0.598 $\pm$ 0.012	0.525 $\pm$ 0.049
COADREAD	0.616 $\pm$ 0.032	0.644 $\pm$ 0.009	0.629 $\pm$ 0.015	0.646 $\pm$ 0.025	0.592 $\pm$ 0.020
HNSC	0.589 $\pm$ 0.034	0.587 $\pm$ 0.012	0.563 $\pm$ 0.031	0.563 $\pm$ 0.013	0.556 $\pm$ 0.027
STAD	0.586 $\pm$ 0.019	0.601 $\pm$ 0.014	0.562 $\pm$ 0.029	0.550 $\pm$ 0.010	0.591 $\pm$ 0.011
Δ vs SurvPath, $n = 25$	+0.039	+0.046	+0.010	+0.020	
95% CI	[-0.016, +0.086]	[+0.004, +0.087]	[-0.019, +0.042]	[-0.031, +0.070]	
t p / W p	0.153 / 0.042	0.045 / 0.042	0.526 / 0.751	0.455 / 0.474	
<i>Integrated Brier score</i>					
BLCA	0.205 $\pm$ 0.008	0.210 $\pm$ 0.004	0.202 $\pm$ 0.007	0.191 $\pm$ 0.002	0.268 $\pm$ 0.031
BRCA*	0.073 $\pm$ 0.000	0.082 $\pm$ 0.004	0.077 $\pm$ 0.002	0.075 $\pm$ 0.002	0.131 $\pm$ 0.005
COADREAD	0.077 $\pm$ 0.006	0.081 $\pm$ 0.003	0.097 $\pm$ 0.001	0.082 $\pm$ 0.002	0.110 $\pm$ 0.009
HNSC	0.221 $\pm$ 0.019	0.215 $\pm$ 0.011	0.205 $\pm$ 0.010	0.200 $\pm$ 0.005	0.341 $\pm$ 0.087
STAD	0.201 $\pm$ 0.006	0.189 $\pm$ 0.009	0.208 $\pm$ 0.013	0.204 $\pm$ 0.001	0.276 $\pm$ 0.043
Δ vs SurvPath, $n = 24$	-0.070	-0.071	-0.068	-0.075	
95% CI	[-0.100, -0.044]	[-0.101, -0.043]	[-0.100, -0.039]	[-0.108, -0.044]	
t p / W p	<0.001 / <0.001	<0.001 / <0.001	<0.001 / <0.001	<0.001 / <0.001	

Table 13: Risk-score correlations of PathQ-Former (20 epochs): Spearman ρ , mean $\pm$ standard deviation over the 15 (fold, seed) pairs, between the fused risk and the same checkpoint’s single-input risks (left) and the other methods’ risks on the same fold and seed (right). Pooled: mean over the 75 pairs.

Fused risk of PathQ-Former vs	BLCA	BRCA	COADREAD	HNSC	STAD	Pooled
own slide-removed risk	0.42 ± 0.13	0.36 ± 0.12	0.53 ± 0.20	0.47 ± 0.08	0.45 ± 0.20	0.45
own RNA-removed risk	0.90 ± 0.08	0.85 ± 0.08	0.73 ± 0.14	0.84 ± 0.09	0.82 ± 0.16	0.83
(slide-removed vs RNA-removed)	0.22 ± 0.07	0.09 ± 0.08	0.16 ± 0.19	0.15 ± 0.10	0.13 ± 0.23	0.15
PathQ-Former w/o aux heads	0.79 ± 0.06	0.62 ± 0.14	0.66 ± 0.09	0.71 ± 0.10	0.69 ± 0.13	0.69
SurvPath	0.63 ± 0.08	0.38 ± 0.17	0.38 ± 0.09	0.45 ± 0.24	0.48 ± 0.12	0.46
ABMIL	0.46 ± 0.14	0.51 ± 0.17	0.38 ± 0.11	0.44 ± 0.13	0.48 ± 0.11	0.46
RNA MLP	0.42 ± 0.09	0.26 ± 0.08	0.37 ± 0.20	0.29 ± 0.06	0.41 ± 0.19	0.35
SNN	0.38 ± 0.11	0.23 ± 0.09	0.36 ± 0.16	0.29 ± 0.07	0.43 ± 0.21	0.34

Table 14: Null code against mean imputation on the same PathQ-Former checkpoints (20 epochs): C-index with one input replaced by the training-mean patch embedding or gene vector, mean $\pm$ sample standard deviation over three seeds of the five-fold mean (as in Table 5; one seed, hence no sd, for the variant without auxiliary heads), and the pooled difference from the complete-input score over the 25 seed-averaged folds with t -test and Wilcoxon p -values. The null-code rows are those of Table 4.

Condition	BLCA	BRCA	COADREAD	HNSC	STAD	Δ	t p	W p
<i>PathQ-Former, mean imputation (three seeds)</i>								
RNA removed	0.597 ± 0.003	0.607 ± 0.035	0.602 ± 0.012	0.589 ± 0.014	0.581 ± 0.018	-0.009	0.29	0.49
Slide removed	0.595 ± 0.027	0.591 ± 0.060	0.580 ± 0.016	0.558 ± 0.024	0.521 ± 0.037	-0.035	0.040	0.045
<i>PathQ-Former w/o aux heads, mean imputation (one seed)</i>								
RNA removed	0.627	0.545	0.631	0.580	0.594	-0.010	0.50	0.83
Slide removed	0.524	0.499	0.600	0.535	0.524	-0.069	3×10^{-4}	2.5×10^{-4}

Table 15: Fusion-block ablations on BLCA under the 10-epoch budget (PathQ-Former without auxiliary heads, seed 0): five-fold mean C-index $\pm$ standard deviation over the five folds with both inputs, and the same checkpoint with the RNA or the slide removed (null codes). The settings are not distinguishable at one seed.

Variant	Both inputs	RNA removed	Slide removed
Base ($K = 32$, fusion depth 2, modality dropout 0.15)	0.609 ± 0.069	0.602	0.630
Modality dropout 0	0.634 ± 0.075	0.609	0.598
Modality dropout 0.3	0.616 ± 0.068	0.624	0.648
Modality dropout 0.5	0.635 ± 0.108	0.587	0.626
$K = 16$ queries	0.638 ± 0.070	0.615	0.615
$K = 64$ queries	0.634 ± 0.127	0.619	0.624
Fusion depth 1	0.598 ± 0.124	0.584	0.611
Fusion depth 3	0.595 ± 0.088	0.585	0.610

C Efficiency

Table 16 reports parameters, latency and peak GPU memory per patient on the first 40 patients of the BLCA fold-0 validation set (medians, one RTX PRO 4000). At 4,096–6,214 patches the two models have equal forward latency; PathQ-Former’s training step takes 104 ms against 176 ms for SurvPath, and its peak memory on full slides is 1.17 GiB against 1.74 GiB. The claim is limited to these slide sizes on this GPU; neither implementation was profiled, and training throughput in practice was bound by reading features from storage.

Table 16: Compute per patient on the first 40 patients of the BLCA fold-0 validation set (medians, one RTX PRO 4000). “All patches” is the median patient (6,214 patches).

Model	Params (M)	Patches	Fwd (ms)	Fwd+bwd (ms)	Peak mem. (GiB)
PathQ-Former, all patches	16.9	6,214	35.0	104.2	1.17
PathQ-Former, 4,096 patches	16.9	4,096	34.7	104.1	0.17
SurvPath, 4,096 patches (as trained)	21.2	4,096	35.7	175.6	0.20
SurvPath, all patches	21.2	6,214	36.0	176.0	1.74

D Pathway Attention

The genomic query block exposes cross-attention from each query over the 275 pathway tokens. Figure 5 shows, per cohort, the eight tokens with the highest mean attention relative to uniform ($1/275$), averaged over validation patients and the five folds (seed 0). The top entries are 3.9 to 18.3 times the uniform weight, but the standard deviation over folds exceeds the mean for most of them (BLCA: apical junction, 6.7 ± 7.9 ; BRCA: glycolysis, 18.3 ± 14.2), so the ranking is descriptive only; attention weights are not a measure of feature importance (Jain & Wallace, 2019).

E Kaplan–Meier Curves

Figure 6 shows disease-specific survival of the high- and low-risk halves of the pooled out-of-fold predictions (20 epochs, seed 0). The log-rank p -values between the halves are, for PathQ-Former, 0.008 (BLCA), 0.054 (BRCA), 0.38 (COADREAD), 0.003 (HNSC) and 0.076 (STAD), and for SurvPath 0.014, 0.035, 0.96, 0.86 and 3×10^{-5} . The pooled curves combine five fold models at one seed and are descriptive.

F Evaluation Pitfalls in Survival MIL Pipelines

The following choices each produce plausible-looking C-index values and are easy to make in a survival MIL pipeline; our pipeline avoids them as stated.

1. **Concordance on discrete bins.** Computing the C-index on the four hazard bins ties a large fraction of the pairs; Harrell’s C-index is computed on continuous follow-up time.
2. **Bin edges fit on the validation fold.** Fitting the quartile edges separately on validation patients puts their labels on a different scale and leaks the validation label distribution; edges are fit on uncensored training patients and reused.
3. **Slide-level samples.** Counting each slide as a sample weights multi-slide patients several times and inflates the apparent cohort size; one sample per patient, with all slides concatenated.
4. **Unscaled gene inputs.** Feeding log-expression values on an arbitrary scale to per-pathway MLPs; genes are min–max scaled with training-split statistics.
5. **Selection on the reported fold.** Selecting a checkpoint on the validation fold that is also the reported fold biases every selection rule upward (Section 4.5); a fixed budget with the final checkpoint and several seeds avoids it.

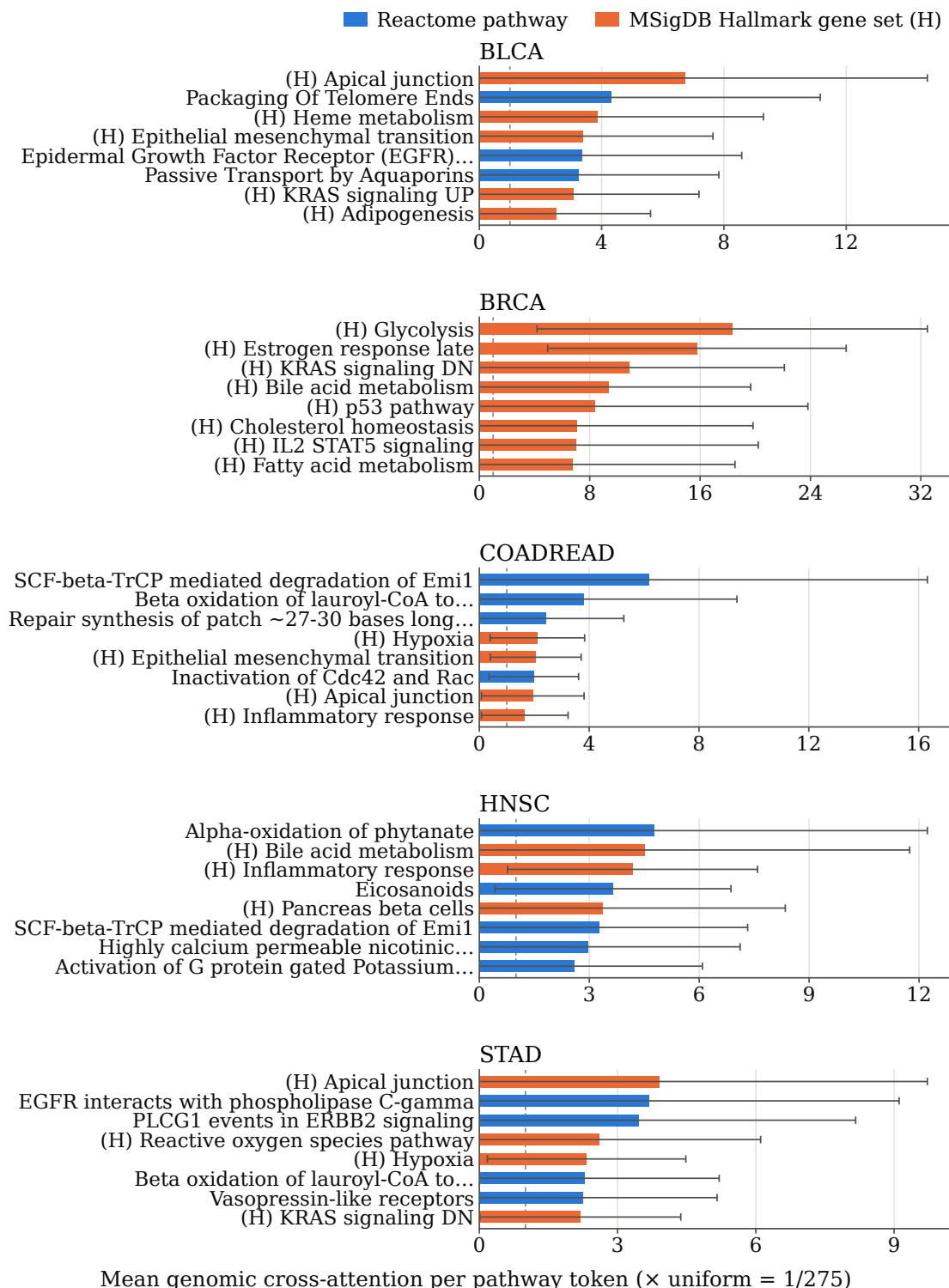


Figure 5: Pathway tokens with the highest mean genomic-query attention per cohort (PathQ-Former, 20 epochs, seed 0), as a multiple of uniform attention over the 275 tokens (dashed line), averaged over validation patients and the five folds. Blue: Reactome pathways; orange, marked (H): MSigDB Hallmark gene sets. Error bars span the mean $\pm$ the standard deviation over the five folds, clipped at zero. Names longer than 45 characters are shortened with an ellipsis.

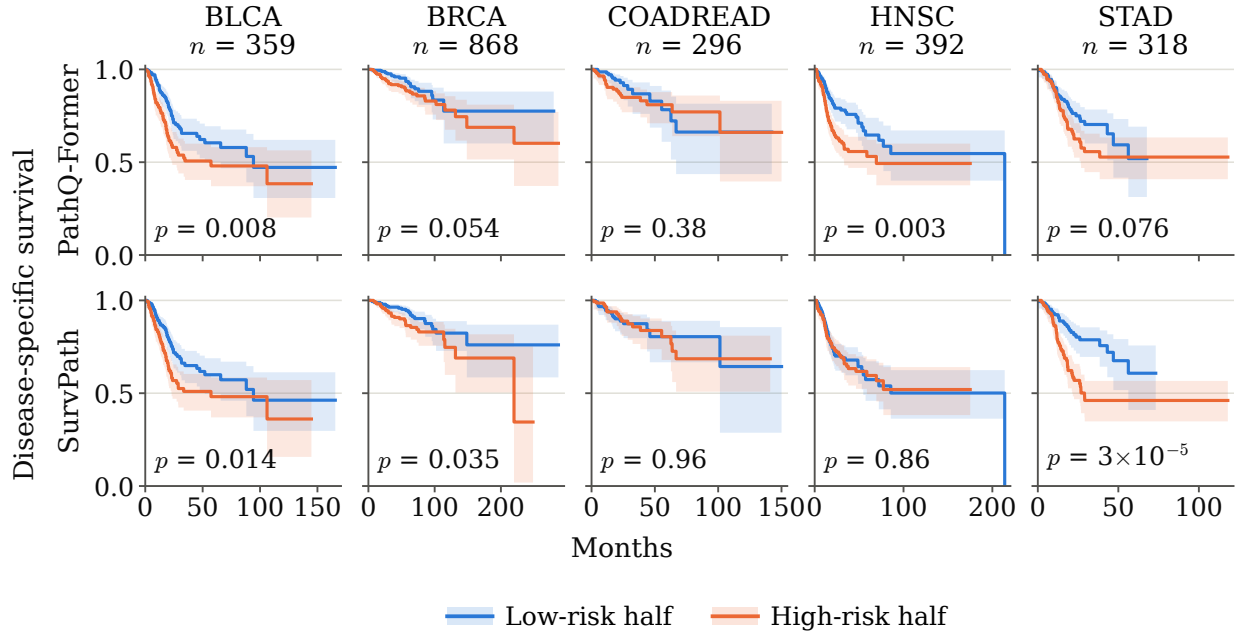


Figure 6: Kaplan–Meier curves of disease-specific survival for the high- and low-risk halves of the pooled out-of-fold validation predictions (20 epochs, seed 0). Top row: PathQ-Former; bottom row: SurvPath; columns: the five cohorts with the pooled number of patients. Within each fold, patients are split at the median predicted risk of that fold’s validation set, and the five folds are then pooled, so every patient appears once. Shaded bands: 95% confidence intervals; p : two-sample log-rank test between the pooled halves.

6. **Non-standard risk and dropout semantics.** Using $1 - S(T)$ as the risk score, or applying modality dropout per batch rather than per sample; risk is $-\sum_t S(t)$ and dropout is per sample.
7. **Extreme times in the Brier-score grid.** A grid that includes the minimum and maximum validation times evaluates on time points at which a single patient is at risk; the interior quartile edges are used.
8. **Untruncated IPCW weights.** Uno’s C-index diverges when the censoring survival function approaches zero at late times; it is truncated at the 75th percentile of training event times.

G Run Inventory and Compute

The archive holds 204 cohort-level runs, each of five folds: 34 under the earlier validation-loss protocol and the single-seed 10-epoch runs of PathQ-Former and SurvPath on five cohorts; 31 BLCA fusion-block ablations, late fusion and 10-epoch seeds 1–2 of PathQ-Former and SurvPath; 128 under the 20-epoch DSS protocol (the nine methods of Table 3 on five cohorts with the seeds listed there); 10 under the OS endpoint (PathQ-Former and SurvPath on five cohorts); and one 20-epoch learning-rate check on BLCA (Appendix A). The validation-loss protocol (patience 5, minimum 6 epochs, 120 folds) selected epoch 1 on median (82% of folds epoch 1–3), trained 7.2 epochs on average, and reported a mean C-index of 0.542 against 0.642 at the best epoch, with 13 of 22 method $\times$ cohort cells within 0.45–0.55; SurvPath on STAD (0.610) and SNN on BRCA (0.572) are among the exceptions. Every trained run stores its results, configuration, per-fold predictions and per-epoch training histories (the late-fusion runs store results and per-fold predictions only); the tables of Section 4 and Appendix B are regenerated from these files by one script. GPU runs were executed on rented RTX PRO 4000/4500 and RTX 2000 Ada GPUs sharing a network volume; throughput was bound by feature I/O rather than by either model.